communications. Pre-shared keys can eliminate this overhead and use more efficient symmetric cryptography. More capable devices can combine PSK cipher suites with ephemeral Diffie-Hellman to achieve forward secrecy.

- IoT communications often need to traversal multiple network hops employing different transport protocols. End-to-end encryption and authentication can be used to ensure that confidentiality and integrity of API requests and responses are not compromised if an intermediate host is attacked. The COSE standards provide similar capabilities to JOSE with better suitability for IoT devices, but alternatives such as NaCl can be simpler and more secure.
- Constrained devices often lack access to good sources of entropy to generate random nonces, increasing the risk of nonce reuse vulnerabilities. Misuse-resistant authentication encryption modes, such as AES-SIV, are a much safer choice for such devices and offer similar benefits to AES-CCM for code size.
- Key distribution is a complex problem for IoT environments, which can be solved through simple key management techniques such as the use of key distribution servers. Large numbers of device keys can be managed through key derivation, and ratcheting can be used to ensure forward secrecy. Hardware security features provide additional protection against compromised devices.

[1] DTLS is limited to securing unicast UDP connections and can't secure multicast broadcasts currently.